

UNIT-II

Development & adoption of low-cost housing technology:

- low-cost housing means constructing the houses with ~~low quality materials~~ low construction costs.
- low-cost housing houses mainly depend on their incomes, without sacrificing the quality of materials (minimum housing ^{costing} requirements).
- Pre-fabrication materials [ready-made materials]
 - ↳ construction can be very fast.
 - ↳ saves money
 - ↳ saves time.
- low-cost housing techniques are mainly depends on 3 factors:
 - 1] structural design
 - 2] Budgeting & cost-cutting
 - 3] materials used.
- professional guidelines for housing [how-to construct the houses with low-cost]
- The government of India has decided to establish the ¹⁹⁷² NBO [National building organization].
 - * Innovative construction techniques.
 - * Improved building materials.
- Pre-cast elements
 - Roofing or flooring
 - Pre-cast doors & window frames.
 - Pre-cast walls.

→ using new construction machines & motors.

- ↳ reduce the labourers.
- ↳ reduce construction time.

→ selection of load-bearing structure instead of frame-structures.

→ Because load-bearing structure can reduce less concrete, cheaper, less steel rods when compared to the framed structure.

↳ easy to construct

↳ load-bearing structures are very flexible.

→ pre cast stair-cas system.

* foundation for low-cost housing tech

→ foundation ^{imp aspect} 10-15% of construction cost.

→ Generally 3-4 ^{feet} foundations in normal soil.

→ 2-feet also enough [saves the more construction cost]

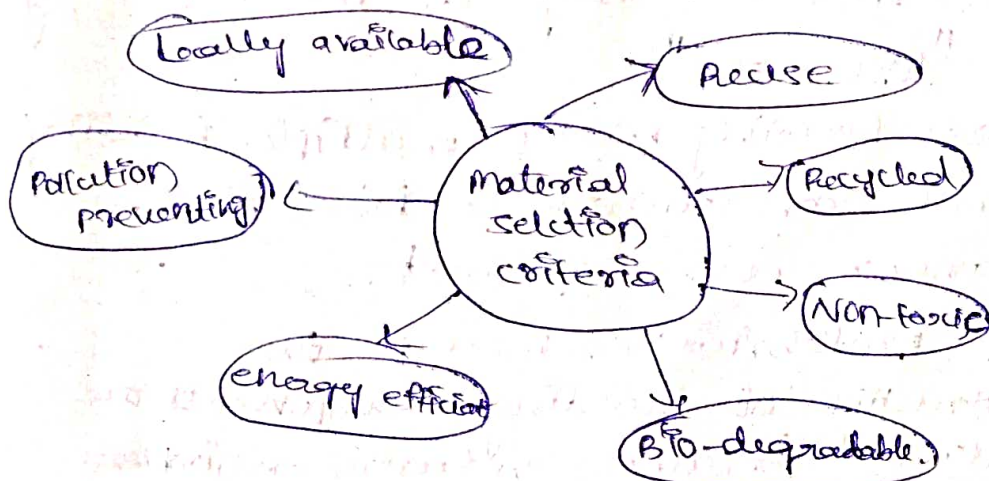
Importance of low-cost housing technology

1) Affordability

2) Improving the living conditions [safety, security,

3) sustainability [reduce the waste, ^{degradability} minimizing environmental impact, Eco-friendly].

4) Economic development



development of low-cost housing techniques

- 1] Pre-fabrication [Residential Buildings
commercial
Infrastructure]
- 2] modular construction.
- 3] Government initiatives.
- 4] Alternate Building materials.
- 5] Energy efficiency.
- 6] community based approaches.

benefits of low-cost housing technique

- 1] cost saving
- 2] time management
- 3] quality control
- 4] Reducing no. of labourers.
- 5] Efficiency.
- 6] Durability.
- 7] Sustainability (eco-friendly, reduce waste).
- 8] safety
- 9] flexibility.

Adoption of total prefabrication of mass housing

- The main advantage of mass housing is "low-cost".
- mass housing many multiple buildings can be constructed in the similar manner with low-cost.
- Total Prefabrication means the construction of building components are controlled by the industry environment.

→ It reduces the on-site work.
benefits of total pre-fabrication in
mass housing

- ① speed construction
- ② cost-efficiency
- ③ time-management
- ④ quality control
- ⑤ sustainability
- ⑥ waste reduces
- ⑦ energy efficiency
- ⑧ durability
- ⑨ flexibility
- ⑩ reducing No. of labourers.
- ⑪ reduces on-site work

considerations for adoption.

-
- ① Investment [requires more]
 - ② Infrastructure
 - ③ Government initiatives.
 - ④ public-private partnerships
 - ⑤ Regulatory framework.
 - ⑥ skilled labourers.
 - ⑦ Transportation challenges.
 - ⑧ sustainability consideration.
 - ⑨ public perception.
 - ⑩ climatic condition.
 - ⑪ maintenance
 - ⑫ connections & joints.

General Remarks on pre-cast roofing / flooring systems

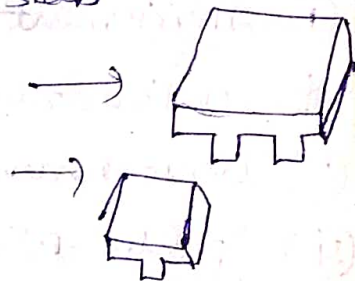
→ pre-cast roofing and flooring systems are construction techniques which are involved the use of pre-cast elements (stairs) premanufactured materials, precast concrete] to create the structural elements of a building roof's and floors.

Advantages :

considerations for challenge : $\begin{cases} \text{Fire resistance} \\ \text{Long Life} \end{cases}$

* Some commonly used types of pre-cast floor and roof slabs include :

- 1] solid flat slabs.
- 2] channel slab
- 3] Hollow core slab
- 4] double tees →
- 5] single tees. →



Economical wall system [diagram]

→ An economical wall systems is a construction method or approach that cost-effective, to meet the essential requirements of the building wall, such as

- * structural integrity
- * insulation
- * aesthetics

→ Achieving the economic wall systems typically involves

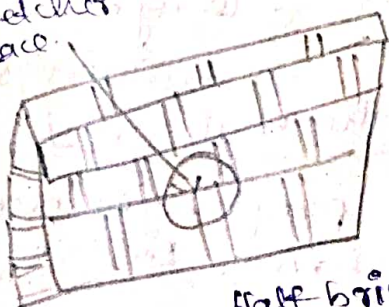
- 1) selecting materials
- 2) techniques
- 3) Designs that can minimize the construction costs without compromising the quality or performance.

⇒ Here some of the strategies for creating an economic wall systems =

- 1] material selection [concrete bricks, bricks, steel rods, slabs]
- 2] standardization [standardization of walls dimension & components can reduce construction labour cost]
- 3] minimize wall thickness
- 4] reduce openings [like doors, windows]
- 5] efficient building techniques, ^{construction}
- 6] pre-fabrication
- 7] modular construction.
- 8] cost Benefit-Analysis
- 9] Local sourcing [reduce transportation costs]
- 10] simplified designs. [straight forward wall designs] can be reduce cost (cost labour cost)
- 11] Energy efficiency.
- 12] Reuse & Recycle.
- 13] maintenance consideration [choose the materials & finishes that are durable].

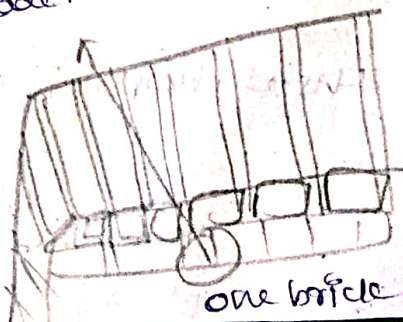
Single brick Thick Loading Bearing wall

single skin brick wall
stretcher face



half brick wall

one brick thick solid wall
header face



one brick thick solid wall

→ standard thickness of single brick thick load bearing wall is 230-250 mm (9-10 inches).

→ The thickness of a brick wall ~~or~~ concrete wall is depends on its height or unsupported height.

→ Red bricks are more stronger than the fly ash bricks.

→ These bricks are more flexible for load bearing applications.

→ A single-brick thick load bearing wall is a type of masonry wall which can support the vertical load of a building structure.

→ This type of wall is manufactured by single row of bricks or single layer of bricks.

Some key points in single-brick thick load bearing wall

- 1] Load bearing
- 2] Thickness (one brick thickness)
- 3] Reinforcement [for extra support ^{steel rods} & prevent from failure]
- 4] Staggered bond [sto.]
- 5] use [older buildings & historical structures]
- 6] maintenance [careful]

Not applicable for taller & larger

7] Insulation [limited thermal and ^{buildings} sound properties due to single-brick thickness].

19 cm thick load bearing masonry wall

- ~~12 cm~~ masonry walls are walls which are made up of bricks & concrete blocks.
- 19 cm thick load bearing masonry walls are type of a vertical loads bearing wall.
- 19 cm (approximately 190 mm) thickness brick wall is one type of masonry wall.
- 19 cm walls are more thicker than single-brick thick load bearing wall.
- These walls have more load bearing capacity.

key aspects of 19 cm thick load bearing wall:

- more stronger than single-brick.
- more Affordable
- more load bearing capacity

1] load bearing

2] Thickness

3] Reinforcement

4] use [

5] maintainance

6] staggered bond

7] Insulation.

8] materials

Half brick thick-load bearing wall

- "Half brick" thick-load bearing wall is a type of masonry wall which can be made by the bricks, concrete bricks and other type of masonry materials.
- Half-brick thick-load bearing wall is constructed by half thickness of the standard ~~brick~~ wall.

→ In a standard brick wall, wall is made up of full thickness which is usually around [7.5 to 8 cm] [75 to 80 mm].

→ half-brick thick load bearing walls are made by bricks ^{are placed in} haroto side, this is typically around of [3.5 - 4 cm].

Key points :-

1] load bearing [w.s.]

2] materials

3] Reinforcement

4] Thickness [3.5 - 4 cm]

5] use

6] maintainance

7] staggard bond

8] Insulation

Fly-ash ^{concrete bricks} gypsum thick for masonry :-

→ Fly-ash gypsum bricks also known as Fly-ash gypsum masonry units.

→ Fly-ash gypsum bricks are type of construction materials which can be used in masonry construction.

→ These bricks are made by the fly-ash and gypsum and other additives.

→ These bricks are durable & stronger and eco-friendly.

key points for fly ash gypsum bricks for masonry :

- 1] materials [fly ash which can be generated by the combustion of coal in power plants].
 - 2] use [mainly used for modern construction, latest buildings].
 - 3] Insulation [These bricks have reasonable thermal properties due to presence of fly ash].
 - 4] properties ~~strengths~~ [stronger & durability].
 - 5] sound insulation [provide some sound insulation].
 - 6] size and shape [these bricks are available in diff shapes (lay)].
 - 7] environmental benefits [these bricks are made by fly ash (recycled)].
 - 8] brick composition [fly ash + Gypsum + fine aggregate + water].
 - 9] load bearing.
- These bricks are used for both load bearing & non-load bearing.

Stone block masonry :

Stone block masonry is a type of construction technique which can be made by the natural stone blocks or cut stone pieces which are used to create the structure of building.

→ This is one of the oldest construction technique.

→ This has a rich history in architecture and civil engineering.

key points :

- ① materials [stone block masonry uses a natural stone blocks, these ^{cutting pieces of} are locally available].

In these stone block masonry technique, common types of stones can include

- ① granite
- ② lime stone
- ③ sandstone
- ④ marble
- ⑤ slate

} choice of stone can be depends on availability on aesthetics.

⑥ uses → It can be used for old building. In modern construction, the stone blocks can be used for aesthetic purposes.

⑦ size & shape ∴ The stone blocks locally available [so they doesn't have fixed size & shape]

→ But when we use the cutted pieces of stones for construction, those stone blocks have fixed size & construction.

⑧ modern Applications ∴

These stone block masonries can be used in

- commercial buildings.
- Institutional buildings.
- Residential buildings.

⑨ Historical significance ∴

→ These stone block masonry has a rich history in architectural & civil engineering.

⑩ Strength & durability.

⑪ Aesthetics [physical appearance of a building]

⑫ maintainance

⑬ Bond pattern

⑭ cutting & shaping

⑮ load bearing & Non-load bearing.

Adoption of Precast (R.C) planks and join system for roof/floor in the building

R.C → Reinforced concrete.

→ Adoption of precast reinforced concrete (R.C) planks and join system for roof/floor in the building has a several advantages in terms of

CE i] construction efficiency

CE ii] cost effectiveness.

iii] overall structure of the building.

iv] time management.

→ pre-cast (R.C) planks are manufactured in off-site and transported to the construction-site.

Here some of the key-points to consider the adoption of precast (R.C) planks —

1] speed of construction

2] durability

3] cost efficiency.

4] Transportation challenges

5] fire resistance

6] sustainability

7] quality control

8] flexibility.

9] maintainance [pre-cast materials does not require more maintainance during their lifespan.

10] skilled labourers [It is important to work with architects, experienced engineers and contractors.

→ In summary there are lot of advantages in terms of cost savings, cost efficiency, overall structure, it requires careful planning.